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A Logistic Regression Model: Predicting Coffee Quality Tier without Cupping

Course: Data Analysis I

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I. Introduction

Coffee is one of the most widely consumed beverages and agricultural exports for many developing countries. A coffee's quality grade (Commercial, Premium, or Specialty) has significant economic consequences for farmers, exporters, importers, and roasters. Being able to predict coffee quality early in the supply chain can help stakeholders make better sourcing decisions, reduce waste, and reward high-quality producers with appropriate pricing.

This project aims to develop and evaluate multiclass classification models that predict the quality tier of Arabica coffee samples using two types of features:

1. Production-level attributes such as growing altitude, processing method, bean variety, and physical defects.
2. Sensory evaluation scores assigned by certified Q-graders during cupping sessions.

By comparing these two feature sets, we aim to determine how well coffee quality tier can be predicted from production-side features alone, without any cupping data, and to explore the implications for real-world applications such as pre-screening, pricing, and farmer self-assessment.

The dataset used in the project is the Coffee Quality Institute (CQI) Arabica Quality Database [LeDoux, 2018], a public dataset that contains 1,311 professionally evaluated Arabica coffee samples from 34 countries, spanning harvest years from 2009 to 2018. Each record includes origin information, processing details, physical defect counts, altitude data, and 10 individual sensory scores (Aroma, Flavor, Aftertaste, Acidity, Body, Balance, Uniformity, Clean Cup, Sweetness, and Cupper Points), each scored on a 1–10 scale by certified Q-graders, which together sum to the Total Cup Points score (maximum 100). From Total Cup Points, we derived our target variable `quality_tier` using the Specialty Coffee Association's standard grading threshold: Commercial (below 80), Premium (80–84), and Specialty (85 and above) [SCA, 2005].

After preprocessing—which primarily includes parsing messy altitude and harvest year fields, handling missing values, and dropping a small number of rows with unparseable data—this dataset contains 1,251 Arabica coffee samples across three different quality tiers: Commercial (13.9%), Premium (78.0%), and Specialty (8.1%).

II. Data Preprocessing

The raw CQI dataset contains 1,311 rows and 44 columns mixing sensory scores, production metadata, administrative identifiers, and pre-computed cleanup fields of uneven quality. This section documents every step of our cleaning process in the order it was performed in the notebook.

II.1 Initial Data Inspection

The dataset was loaded from the CQI GitHub repository using `pd.read_csv()`, returning 1,311 rows and 44 columns. All column names were converted to `snake_case` format to ensure consistency throughout the analysis. A column-level summary was then computed using `df.dtypes`, `df.nunique()`, and `df.isnull().sum()` to capture data types, cardinality, and missingness for every column.

Calling `df.duplicated().sum()` confirmed that there were no duplicate rows. However, many columns had missing values, which exhibited a clear pattern: the columns with the highest proportion of missing values were administrative identifiers, namely `lot_number` (79.4% missing), `farm_name` (27.2%), `mill` (23.7%), `producer` (17.5%), and `company` (15.9%). This pattern is consistent with the CQI submission process, where producers may submit coffees without complete administrative paperwork.

Among coffee-attribute columns, missingness appeared in `color` (20.4%), the three altitude-related fields (each 17.3% missing), the raw altitude column (17.0%), `variety` (15.3%), and `processing_method` (11.6%). Each of these required individual handling decisions in subsequent preprocessing steps.

Most data types were appropriate (the ten sensory scores stored as `float64`, defect counts as `int64`), but two columns stored as object required custom parsing: `altitude` (mixed unit formats, e.g., “1500 m”, “4200 feet”, “1000–1500m”), and `harvest_year` (inconsistent formatting with year ranges, quarter notation, and multilingual month names). The pre-computed altitude cleanup columns (`altitude_low_meters`, `altitude_high_meters`, and `altitude_mean_meters`) appeared clean as `float64` but, as discovered later in Section II.4, contained systematic parsing bugs.

II.2 Drop Irrelevant Columns

Based on the inspection in Section II.1, 23 columns were identified as not useful for predicting coffee quality and were removed before further analysis. These fell into four categories:

Zero-variance column. The `species` column contained only one unique value “Arabica” for all rows. Since the dataset is restricted to Arabica coffee, this column is useless and was dropped.

Administrative identifiers. Thirteen columns recorded paperwork metadata: `unnamed:_0` (a stale row-index column), `lot_number`, `ico_number`, `farm_name`, `mill`, `producer`, `company`, `owner`, `owner_1`, `in_country_partner`, `certification_body`, `certification_address`, and `certification_contact`. Including these columns in the model would force it to memorize specific producers or certification bodies rather than learning generalizable patterns about coffee quality.

Process metadata. Four columns recorded certification logistics or batch information rather than coffee properties: `grading_date` and `expiration` describe when each sample was processed, while `bag_weight` and `number_of_bags` describe the size of the submission batch.

Redundant or buggy columns. Five columns either duplicated information or contained errors. The four altitude-related columns (`unit_of_measurement`, `altitude_low_meters`, `altitude_high_meters`, and `altitude_mean_meters`) were the attempted cleanup of the raw altitude text but contained systematic parsing bugs (see Section II.4). The `region` column had 343 unique country region names with most having only 1–2 samples; geographic information was instead captured through `country_of_origin`, later regrouped into broader continental regions in Section II.9.

After this drop, the working dataset contained 1,311 rows and 21 columns.

II.3 Initial Processing Decisions

After dropping irrelevant columns, missingness was re-examined across the remaining 21 columns. The missing pattern persisted: `color` (267 missing, 20.4%), `raw altitude` (223 missing, 17.0%), `variety` (201 missing, 15.3%), `processing_method` (152 missing, 11.6%), and `harvest_year` (47 missing, 3.6%), with single missing values in `country_of_origin` and `quakers`.

One row with `total_cup_points == 0` was dropped as a data-entry error (the practical minimum cupping score is around 60), leaving 1,310 rows.

The classification target `quality_tier` was constructed from `total_cup_points` using the SCA grading thresholds [SCA, 2005]:

- **Commercial:** total cup points below 80
- **Premium:** total cup points between 80 and 84.99
- **Specialty:** total cup points 85 or above

Applied to the data, the target distribution was 174 Commercial samples (13.3%), 1,033 Premium samples (78.9%), and 103 Specialty samples (7.9%). This class imbalance shaped several decisions in model development (Section IV).

II.4 Altitude Parser

The `altitude` column required the most preprocessing effort. The four altitude cleanup columns provided in the dataset contained systematic bugs that made them unreliable, so a customized parser was developed using the raw altitude text column.

Inspection of `altitude_mean_meters` revealed impossible values: the maximum was 190,164 m (for reference, Mount Everest's summit is 8,849 m). Two problem categories were identified:

- **Decimal-stripping errors:** Row 1144 had a raw altitude of “1901.64” (Guatemala) but was stored as 190,164.0; row 1040 had “1100.00 mosl” (Nicaragua) stored as 110,000.0. In both cases, the parser stripped the decimal point and treated the digits as a single integer.
- **Producer-side reporting errors:** Several Guatemalan rows reported “3280” with `unit_of_measurement = 'm'`, likely confusing feet with meters; multiple Myanmar rows reported altitudes between 3,800 and 4,287 m, which are biologically impossible; one Brazilian row reported “11000 metros”.

A `parse_altitude()` function was written with four steps:

1. Extract all numeric values using the regular expression `\d+\.\d*` (capturing both integers and decimals, preventing the decimal-stripping bug).
2. If two numbers are extracted (e.g., “1000–1500m”), use the midpoint.
3. If the string contains *feet*, *ft*, or an apostrophe, convert from feet to meters using the factor 0.3048.
4. If the resulting altitude falls outside the biologically plausible range of 100–3000 m, flag the value as missing.

After parsing, 948 rows had valid altitudes and 360 rows were flagged. Missing altitudes were imputed with the median altitude of valid samples from the same `country_of_origin` to preserve real geographic signal. Two rows whose `country_of_origin` was missing could not be imputed and were dropped, leaving 1,308 rows.

The final `altitude_imputed_m` column has a mean of 1,301 m, median of 1,350 m, and standard deviation of 398 m, consistent with global Arabica production.

II.5 Harvest Year Parser

The `harvest_year` column was stored as object because of inconsistent submission formats. The 46 unique values fell into five formatting patterns: plain years (e.g., “2012”), year ranges (e.g., “2013/2014”), two-digit year codes (e.g., “08/09 crop”), quarter notation (e.g., “1T/2011”), and date text in multiple languages (English and Spanish), including junk entries such as “TEST” and “mmm”.

A `parse_harvest_year()` function was written using two regex-based strategies:

1. Find any 4-digit year in the range 2000–2029 using `\b(20[0-2]\d)\b`. If multiple years are found, take the later year.
2. Fallback for two-digit codes: search for 2-digit codes following a slash using `/(\d{2})\b` (e.g., “4T/10” → 2010).
3. Strings with no extractable year information return NaN.

The parser successfully extracted years for 1,251 of 1,308 rows. The 57 unparseable rows were dropped. The cleaned `harvest_year_clean` column spans 2009 to 2018 with a median of 2014 and standard deviation of 1.85 years.

II.6 Variety Grouping

The `variety` column contained 29 unique values plus 156 missing entries (12.5% missingness). To reduce cardinality while preserving meaningful information, varieties were grouped into 10 botanical families using the Arabica variety taxonomy maintained by World Coffee Research and the Specialty Coffee Association:

- **Bourbon:** Bourbon and Yellow Bourbon.
- **Typica:** Typica, Mundo Novo, Blue Mountain, and Java.
- **SL_Kenyan:** SL14, SL28, and SL34.
- **Specialty:** Geisha, Pacamara, Pacas, Marigojipe, Peaberry, and Moka Peaberry (rare specialty varieties, each with fewer than 15 samples).
- **Caturra, Catuai, Catimor, Hawaiian Kona:** Kept as their own groups (each with at least 20 samples and a distinct cultivation identity).
- **Other:** Remaining low-sample varieties.
- **Unknown:** The 156 missing entries, preserved as their own informative category.

The final distribution: Bourbon (259), Caturra (252), Typica (247), Unknown (156), Other (127), Catuai (72), Hawaiian Kona (44), SL_Kenyan (38), Specialty (36), and Catimor (20).

II.7 Processing Method Cleanup

The `processing_method` column had five unique values plus 100 missing entries (8.0% missingness). Slash-notation labels were shortened (e.g., “Washed / Wet” → “Washed”), and the 12-sample “Pulped natural / honey” category was merged into “Semi-washed” due to process similarity. Missing entries were labelled “Unknown”.

Final distribution: Washed (808, 64.6%), Natural (249, 19.9%), Unknown (100, 8.0%), Semi-washed (68, 5.4%), and Other (26, 2.1%).

II.8 Color Cleanup

The `color` column contained three unique values plus 220 missing entries (17.6% missingness): Green (839, 67.1%), Bluish-Green (112, 9.0%), and Blue-Green (80, 6.4%). The distinction between “Bluish-Green” and “Blue-Green” is linguistic rather than meaningful; both describe freshly-milled, high-moisture green coffee beans. They were merged into a single “Blue-Green” category. Missing entries were labelled “Unknown”.

Final distribution: Green (839, 67.1%), Unknown (220, 17.6%), Blue-Green (192, 15.4%).

II.9 Country to Region Grouping

The `country_of_origin` column contained 34 unique values with severe imbalance (the top five countries accounted for 62% of samples, while 15 countries had fewer than 10 samples each). Countries were grouped into five coffee-growing regions based on the International Coffee Organization’s geographic classifications:

- **Central America:** Mexico, Guatemala, Honduras, Costa Rica, Nicaragua, El Salvador, Panama.
- **South America:** Colombia, Brazil, Peru, Ecuador.
- **East Africa:** Ethiopia, Tanzania, Uganda, Kenya, Malawi, Burundi, Rwanda, Zambia, and Côte d’Ivoire.
- **Southeast Asia:** Taiwan, Thailand, Indonesia, China, Myanmar, Vietnam, Philippines, Laos, and Papua New Guinea.
- **Caribbean/US:** Hawaii, Haiti, Puerto Rico, U.S. mainland, and Mauritius.

Final distribution: Central America (561, 44.8%), South America (309, 24.7%), Southeast Asia (161, 12.9%), East Africa (147, 11.8%), and Caribbean/US (73, 5.8%).

II.10 Moisture Cleanup

The `moisture` column appeared complete but 229 rows (18.3%) had a moisture reading of exactly 0, which is physically impossible for coffee beans (target range: 10–12% for shelf stability). These were interpreted as hidden missingness.

Regional medians for moisture (computed from non-zero values) had a spread of only 0.01 percentage point across all five regions, so the simpler global median of 0.110 (11.0%) was used to fill all 229 zero values.

After imputation: minimum 0.01, median 0.11, maximum 0.28, standard deviation 0.018.

II.11 Quakers Missing Value

The quakers column contained a single missing value, imputed with the mode 0, since over 92% of samples were already at zero. Its effect on summary statistics and model development is negligible at a sample size of 1,251.

II.12 Drop color_group

Chi-square tests of independence between each cleaned categorical feature and quality_tier (full results in Section III.8) yielded the following ranking: region_group (188.98) > variety_group (126.41) > processing_group (47.78) > color_group (42.08).

color_group showed the weakest association, the highest missingness of any retained categorical feature (17.6%), and only three categories (one being a missing-data placeholder). It was therefore dropped from the feature set. The working dataset after this step contained **1,251 rows and 20 columns**.

III. Exploratory Data Analysis

This section explores the cleaned dataset to understand how production features and sensory scores relate to the quality tier.

III.1 Target Distribution

The dataset is highly imbalanced:

- Premium: 78.0%
- Commercial: 13.9%
- Specialty: 8.1%

To address this imbalance, class_weight='balanced' was applied in one of the logistic regression models (Model A – balanced) to ensure better performance across all classes.

III.2 Numeric Production Features

Feature	Count	Mean	SD	Min	25%	Median	Max
altitude_imputed_m	1,251	1,310.7	390.0	100.0	1,054.5	1,350.0	2,560.0
harvest_year_clean	1,251	2,013.7	1.8	2,009	2,012	2,014	2,018
moisture	1,251	0.111	0.018	0.01	0.11	0.11	0.28
category_one_defects	1,251	0.42	1.84	0	0	0	31
category_two_defects	1,251	3.66	5.41	0	0	2	55
quakers	1,251	0.17	0.80	0	0	0	11

Table 1: Descriptive Statistics for Numeric Production Features

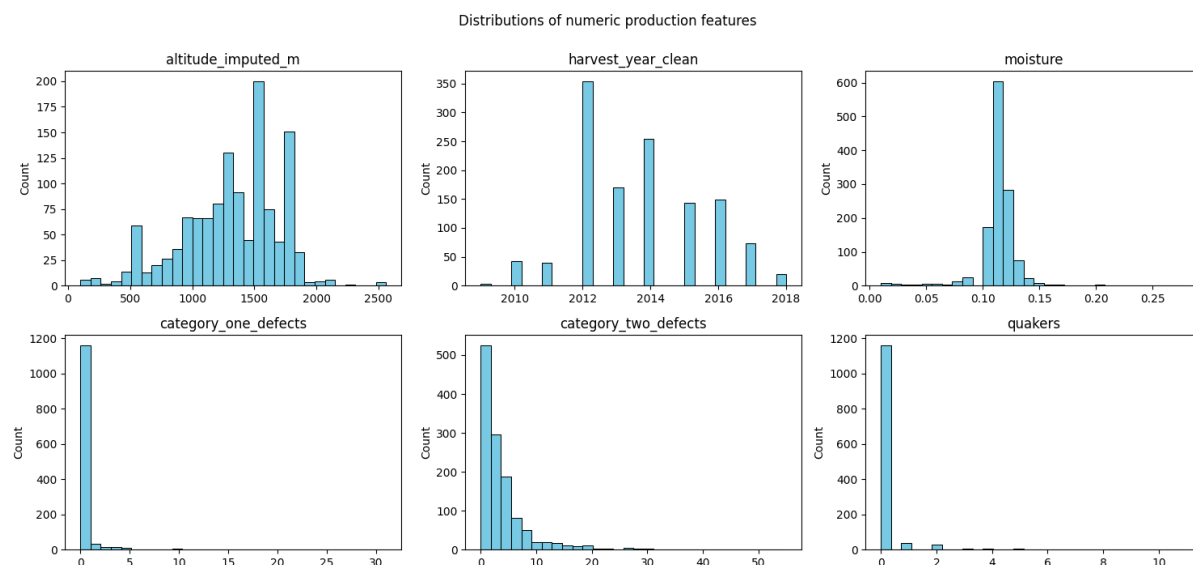


Figure 1: Distribution of Numeric Production Features

Key observations:

- **Altitude** (`altitude_imputed_m`): The average altitude is 1,311 m, with most samples between 1,054 m and 1,600 m.
- **Harvest year** (`harvest_year_clean`): Samples range from 2009 to 2018, with most collected between 2012 and 2015.
- **Moisture**: The average moisture level is about 11.1%, consistent with standard industry levels.
- **Category one defects**: Most coffees have no primary defects (median = 0); the distribution is heavily right-skewed (maximum = 31).
- **Category two defects**: Similarly right-skewed (maximum = 55, average 3.66).
- **Quakers**: Most samples have zero quakers (median = 0, mean = 0.17). Due to low variation and weak correlation with the target, `quakers` and `harvest_year_clean` were excluded from the final model.

III.3 Sensory Scores

Feature	Count	Mean	SD	Min	25%	Median	Max
aroma	1,251	7.57	0.32	5.08	7.42	7.58	8.75
flavor	1,251	7.52	0.34	6.08	7.33	7.50	8.83
aftertaste	1,251	7.40	0.35	6.17	7.25	7.42	8.67
acidity	1,251	7.54	0.32	5.25	7.33	7.50	8.75
body	1,251	7.52	0.29	5.25	7.33	7.50	8.58
balance	1,251	7.52	0.35	6.08	7.33	7.50	8.75
uniformity	1,251	9.85	0.49	6.00	10.00	10.00	10.00
clean_cup	1,251	9.84	0.74	0.00	10.00	10.00	10.00
sweetness	1,251	9.91	0.45	1.33	10.00	10.00	10.00
cupper_points	1,251	7.50	0.43	5.17	7.25	7.50	8.67

Table 2: Descriptive Statistics for Sensory Score Features

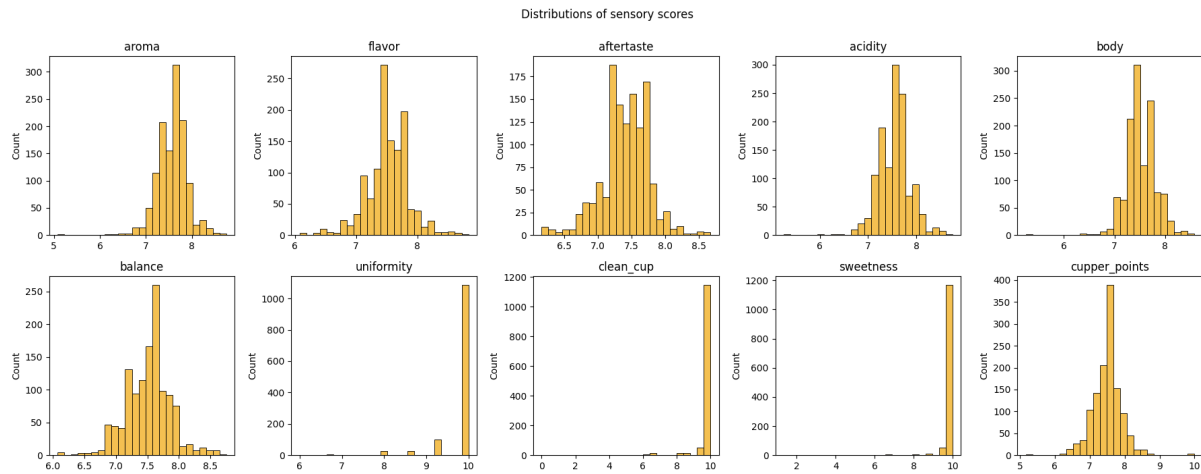


Figure 2: Distribution of Numeric Sensory Features

The ten sensory scores divide into two groups:

- **Analytical scores** (aroma, flavor, aftertaste, acidity, body, balance, cupper_points): mostly fall between 7.0 and 8.5, with standard deviations from 0.29 to 0.43. These scores vary across samples, making them useful for distinguishing quality levels.
- **Protocol scores** (uniformity, clean_cup, sweetness): generally very high, often close to 10, with mean values above 9.85 and standard deviations between 0.45 and 0.74. Since most samples receive similarly high scores, these features are less useful for distinguishing quality levels.

III.4 Categorical Features

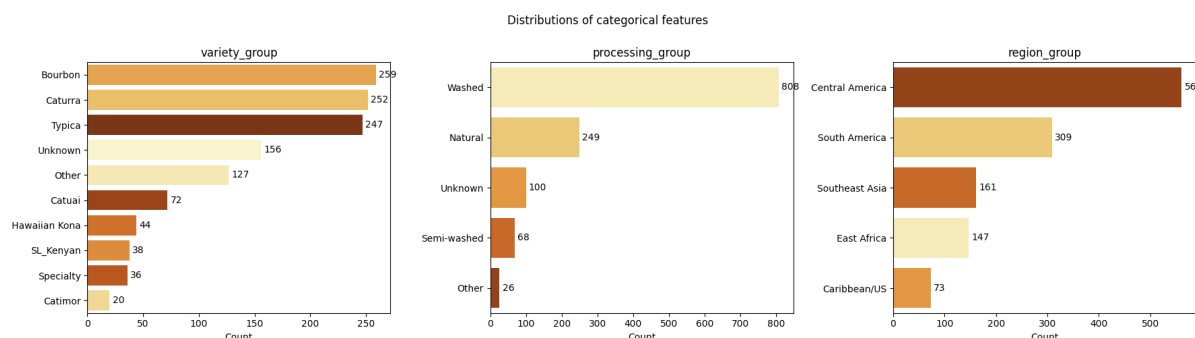


Figure 3: Distribution of Categorical Features

- **Variety** (variety_group): Bourbon (259), Caturra (252), and Typica (247) make up around 60% of the dataset.
- **Processing method** (processing_group): Washed (808) dominates; Natural (249), Unknown (100), Semi-washed (68), and Other (26).
- **Region** (region_group): Central America (561), South America (309), Southeast Asia (161), East Africa (147), and Caribbean/US (73).

III.5 Numeric Features vs. Quality Tier

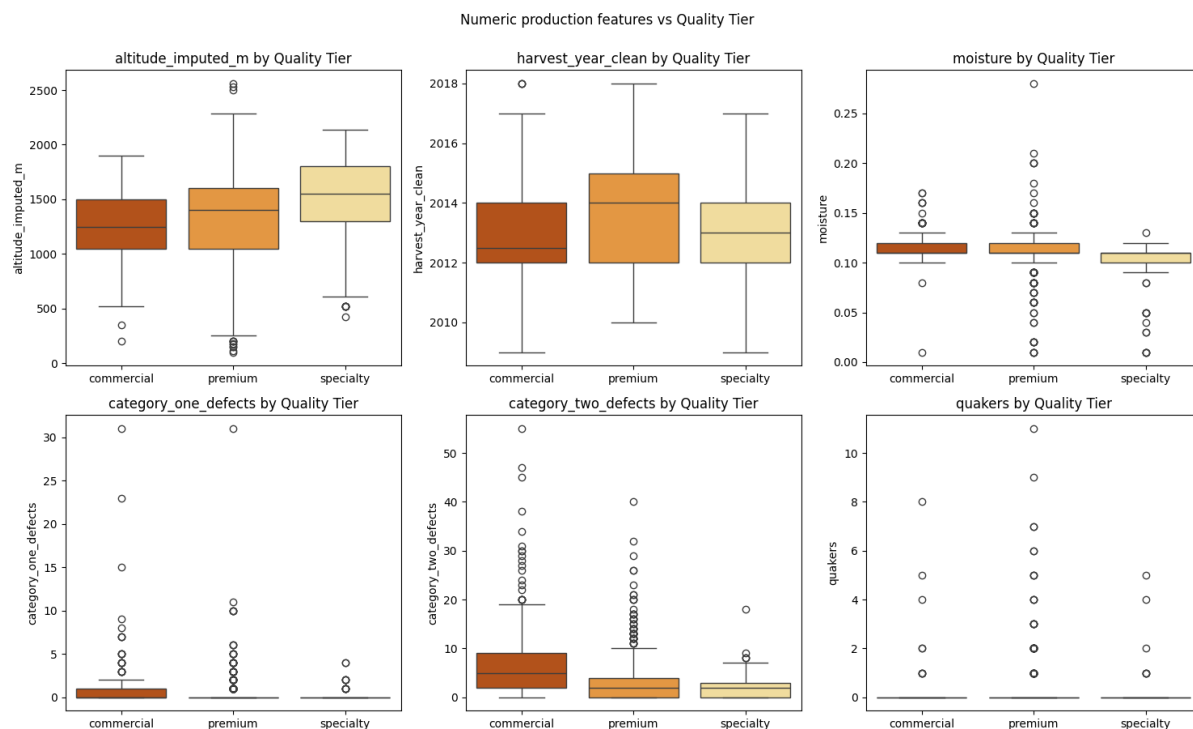


Figure 4: Numeric Production Features vs. Target Group

The boxplots show that:

- **Altitude:** Altitude increases as coffee quality improves; Specialty is grown at higher altitudes, consistent with the view that altitude slows bean growth and improves flavor development [Wintgens, 2009].
- **Category one defects:** Commercial coffee shows a wider distribution; Premium and Specialty coffees have very low and similar defect levels.
- **Category two defects:** Commercial coffee shows higher defect levels; the difference between groups is clearer than for category one defects.
- **Moisture, harvest year, and quakers:** All quality groups show very similar distributions with no clear differences, confirming these features do not help in predicting quality.

III.6 Categorical Features vs. Quality Tier

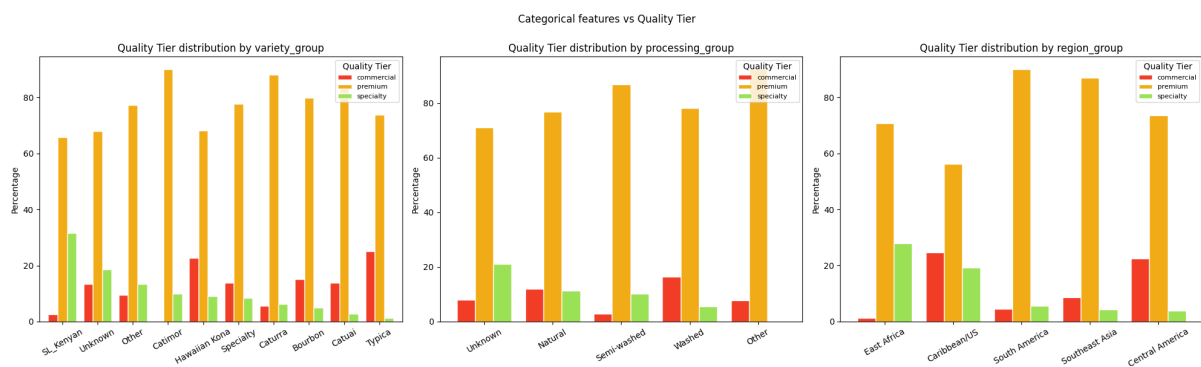


Figure 5: Categorical Features vs. Target Group

The grouped bar plots show:

- **Variety:** SL_Kenyan has the highest proportion of Specialty coffee; Typica has the lowest Specialty rate and the highest Commercial share.
- **Processing method:** Natural-processed coffees tend to produce more Specialty coffee, possibly because natural processing enhances flavor complexity. Semi-washed coffees are mostly Premium.
- **Region:** East Africa has the highest proportion of Specialty coffee. Caribbean/US also shows a relatively high Specialty rate. Southeast Asia has the lowest Specialty rate and a higher share of Commercial coffee.

III.7 Correlation Heatmap

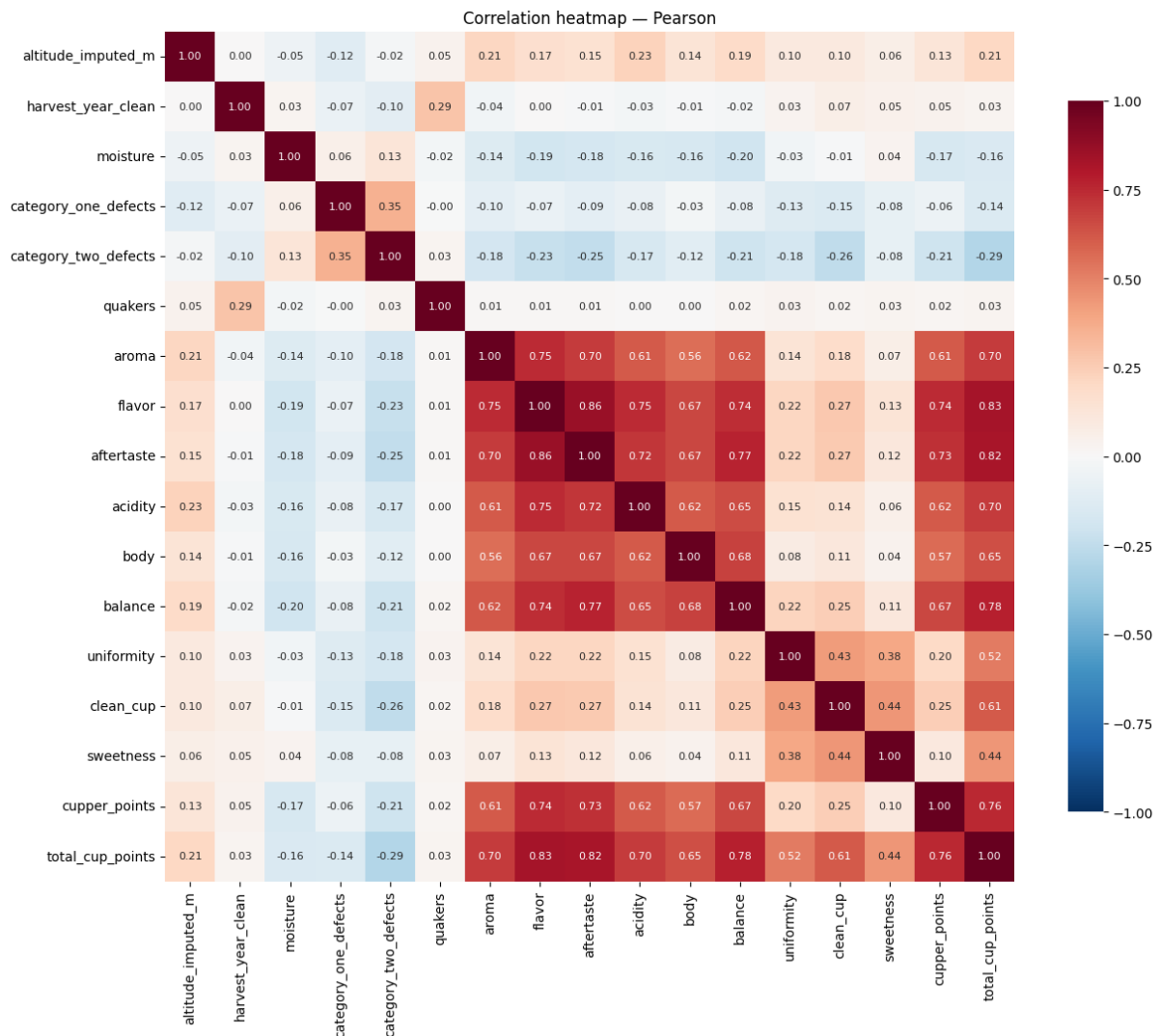


Figure 6: Correlation Heatmap — Pearson

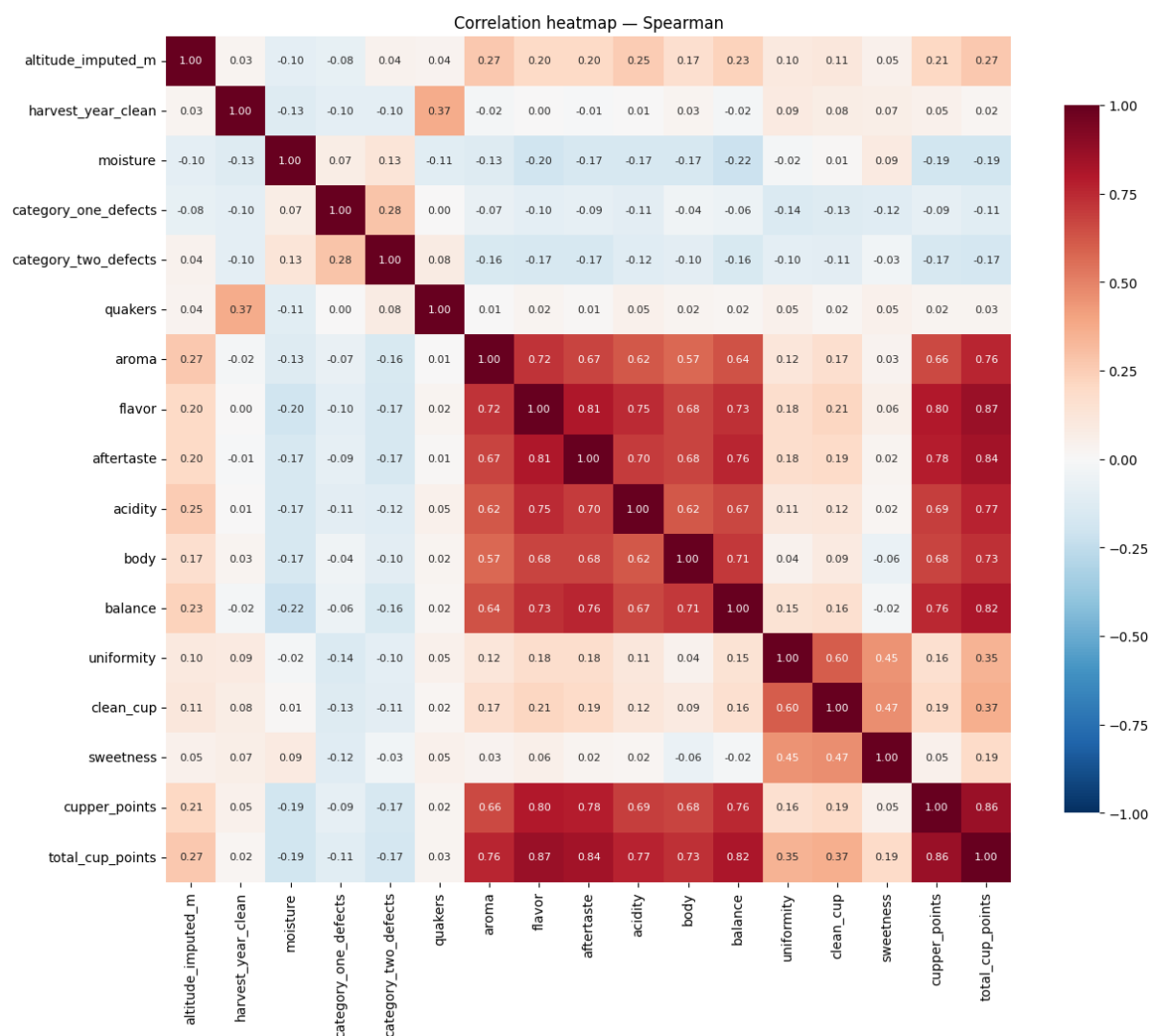


Figure 7: Correlation Heatmap — Spearman

Key findings from the Pearson correlation heatmap:

- **Sensory scores are strongly correlated:** The strongest pairwise correlations are flavor–aftertaste ($r = 0.86$), aftertaste–balance ($r = 0.77$), and aroma–flavor ($r = 0.75$). To avoid multicollinearity, Model B uses only flavor as a representative feature.
- **Flavor is the strongest predictor of total_cup_points:** Flavor ($r = 0.83$), aftertaste ($r = 0.82$), and balance ($r = 0.78$) have the highest correlations with the target. Protocol scores (uniformity, clean_cup, sweetness) show weaker relationships ($r = 0.44$ – 0.61) and were excluded from the model.
- **Production features show weak linear relationships:** The strongest production-related correlations are category_two_defects ($r = -0.29$) and altitude ($r = 0.21$). Other production features have little to no linear correlation with the target.

Spearman results are broadly consistent with Pearson. Notable differences: altitude rises from 0.21 to 0.27 (suggesting mild non-linearity), while category_two_defects drops in magnitude from -0.29 to -0.17 (suggesting Pearson was inflated by extreme defect outliers). Harvest

year and quakers show near-zero correlations under both methods, confirming they should be excluded.

III.8 Chi-Squared Tests

Chi-square tests were used to assess the association between each cleaned categorical feature and `quality_tier`.

Feature	χ^2	df	p-value
region_group	188.98	8	1.34×10^{-36} ***
variety_group	126.41	18	2.57×10^{-18} ***
processing_group	47.78	8	1.09×10^{-7} ***
color_group	42.08	4	1.60×10^{-8} ***

*** $p < 0.001$

Table 3: Chi-Square Test Results for Categorical Features vs. Quality Tier

All four features showed statistically significant associations at $p < 0.001$. The chi-square statistic was used as a relative ranking of association strength:

- `region_group` is the strongest predictor, consistent with the visual pattern in Section III.6 that East Africa shows the highest Specialty rate and Southeast Asia the lowest.
- `variety_group` also shows a strong association, matching the evidence that SL_Kenyan varieties dominate the Specialty tier.
- `processing_group` shows a weaker but significant association.
- `color_group` has the weakest association and was removed from the feature set (see Section II.12).

IV. Model Development

Based on the EDA findings in Section III, logistic regression was selected as the modelling algorithm. It is interpretable, appropriate for multi-class classification, and well-suited to the relatively small, mixed-type feature set. Four models were built to allow structured comparisons.

IV.1 Data Splitting

The dataset of 1,251 rows was split into 80% training (1,000 rows) and 20% test (251 rows) sets using stratified sampling (`stratify = y`) to maintain class proportions, with `random_state = 42` for reproducibility.

Split	Total	Commercial	Premium	Specialty
Training (80%)	1,000	139 (13.9%)	780 (78.0%)	81 (8.1%)
Test (20%)	251	35 (13.9%)	196 (78.1%)	20 (8.0%)

Table 4: Train/Test Split with Class Distribution

IV.2 Model Selection & Training

Four models were trained:

- **Dummy Classifier:** Used as a baseline by always predicting “Premium”. Achieved 78.1% accuracy but a macro-F1 of 0.29, confirming it completely fails on Commercial and Specialty classes.
- **Model A – Unbalanced:** Logistic regression on production features with default settings, tending to favor the dominant Premium class.
- **Model A – Balanced:** Logistic regression on production features with `class_weight='balanced'`, giving more weight to minority classes. This was selected as the primary Model A due to better minority-class performance.
- **Model B – Flavor Only:** Logistic regression trained on the single `flavor` score (Pearson $r = 0.83$ with `total_cup_points`). Serves as a benchmark for sensory-based prediction.

V. Results and Evaluation

V.1 Performance Metrics

All models were evaluated on Accuracy, Macro-F1, Precision, and Recall. Due to class imbalance, Macro-F1 is the primary metric, as it assigns equal weight to all classes and prevents the majority class (Premium) from dominating the evaluation.

Model	Accuracy	Macro F1	Precision	Recall
Dummy Classifier	0.781	0.292	0.260	0.333
Model A – Unbalanced	0.785	0.409	0.615	0.397
Model A – Balanced	0.526	0.463	0.475	0.644
Model B – Flavor Only	0.912	0.832	0.909	0.788

Table 5: Model Performance Comparison

V.2 Model Comparison

- **Dummy Classifier:** High accuracy (78.1%) but very low Macro-F1 (0.292) confirms that accuracy alone is misleading under class imbalance.
- **Model A – Unbalanced:** Slightly better than the dummy baseline (Macro-F1 = 0.409), but still favors the dominant Premium class. This suggests that production features alone provide limited predictive power when class imbalance is not addressed.
- **Model A – Balanced:** Accuracy decreases to 52.6% while Macro-F1 improves to 0.463 and Recall rises to 0.644, indicating better detection of Commercial and Specialty samples at the expense of overall accuracy.
- **Model B – Flavor Only:** Best overall performance: 91.2% accuracy, Macro-F1 of 0.832, Precision 0.909, and Recall 0.788. A single sensory feature substantially outperforms all production-feature models.

V.3 Confusion Matrix Analysis

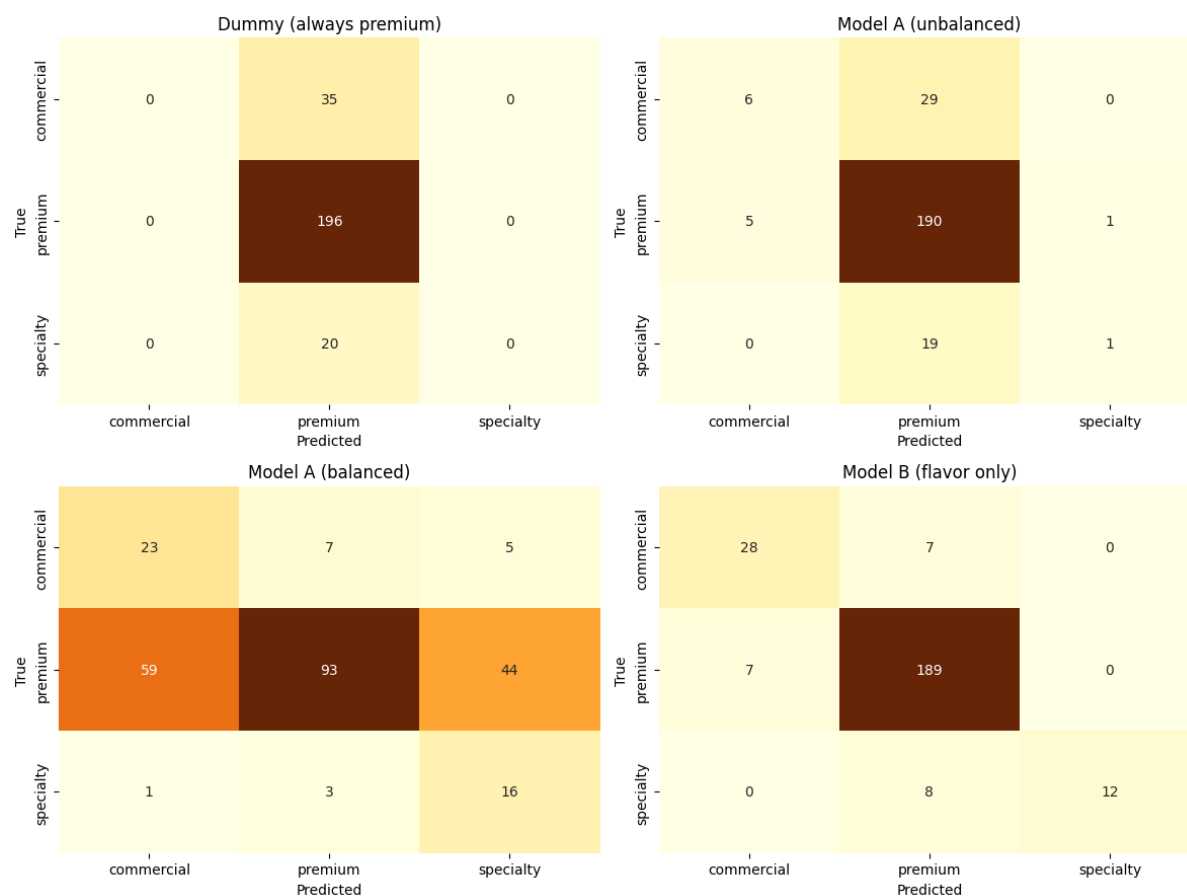


Figure 8: Confusion Matrices

- **Dummy:** Predicts every sample as “Premium”; completely fails on Commercial and Specialty.
- **Model A – Unbalanced:** Correctly identifies most Premium samples but misclassifies nearly all Specialty and Commercial samples.
- **Model A – Balanced:** Identifies more Specialty coffee but still struggles with Commercial due to low precision.
- **Model B:** Mostly correct predictions with only slight confusion between Premium and Specialty.

V.4 Feature Importance

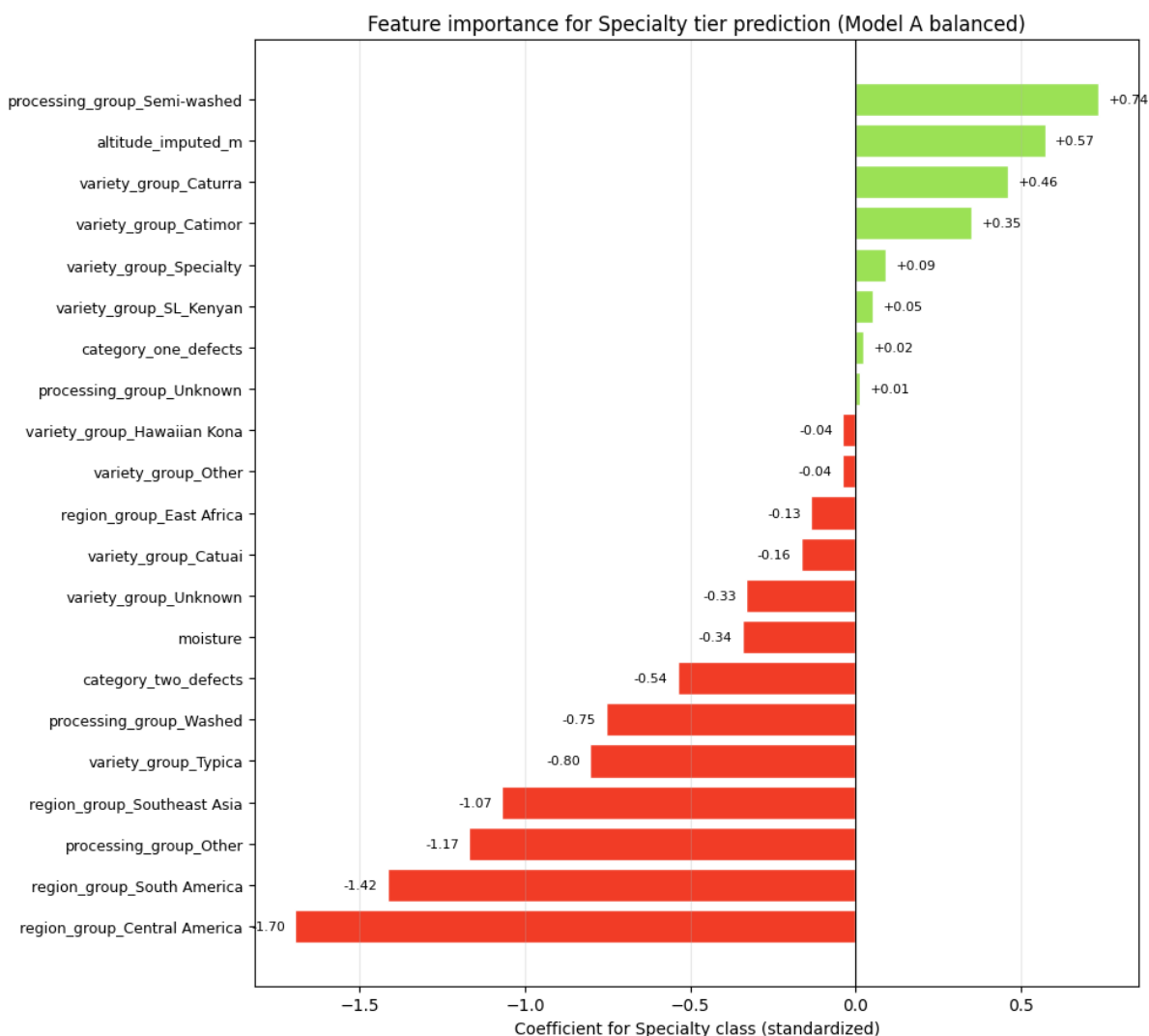


Figure 9: Feature Importance for Model A Balanced

Coefficients for the Specialty class from the balanced logistic regression model:

Strongest positive predictors:

- `processing_group_Semi-washed` (+0.74): strongest positive predictor of Specialty classification.
- `altitude_imputed_m` (+0.57): higher altitude increases the probability of Specialty classification.
- `variety_group_Caturra` (+0.46) and `variety_group_Catimor` (+0.35): both positively associated with Specialty relative to the Bourbon reference.

Notably, `region_group_East_Africa` and `variety_group_SL_Kenyan` do not show strong positive coefficients despite their visual dominance in the bivariate plots (Section III.6). This is due to multicollinearity: most SL_Kenyan samples originate from East African countries, so the model splits the predictive signal between the two related dummy variables. Additionally,

the Caribbean/US reference region has a high Specialty rate (driven by Hawaiian Kona), which reduces the apparent East Africa coefficient.

Strongest negative predictors: `region_group_Central_America` (−1.70) and `region_group_South_America` (−1.42) indicate that coffees from these regions are much less likely to be classified as Specialty compared to the Caribbean/US reference. `processing_group_Other` (−1.17) and `variety_group_Typica` (−0.80) also showed strong negative effects. Defect counts reduce the likelihood of Specialty classification, consistent with industry expectations.

VI. Conclusion & Business Implications

This project shows that production-related information—origin, variety, processing method, altitude, and defect counts—can help predict coffee quality levels. The balanced logistic regression model (Model A) performs substantially better than a simple baseline, proving that it is possible to estimate coffee quality from production-side features alone.

However, the gap between the production-feature model and the single sensory-score reference is large. Model A achieves a Macro-F1 of approximately 0.46 compared to approximately 0.83 for the flavor-score model (Model B), demonstrating that cupping data provides much stronger predictive value than any combination of production features in this dataset.

Business Implications:

- **For buyers and importers:** Coffee from East Africa, especially when semi-washed or naturally processed and with low defects, is often a strong indicator of high quality, allowing buyers to focus on the most promising samples and reduce cupping time and costs [Iburu Coffee, 2021, Ebru Coffee Co., 2025].
- **For farmers and cooperatives:** Lowering defect levels has the greatest effect on improving coffee grade. Better sorting and post-harvest processing tools can greatly increase the chance of reaching higher quality categories [SCA, 2005, WEF, 2020].
- **For quality graders and certifiers:** A simple screening model using basic origin and production data can reduce the need for full cupping while maintaining reliable classification results [Perfect Daily Grind, 2024].

Limitations:

- The dataset is small and has very few Commercial coffee samples, which may reduce model accuracy.
- Logistic regression assumes simple relationships; more advanced models might perform better.
- Some values (e.g., altitude) were estimated using country-level medians, potentially reducing accuracy.

Future Work:

- Adding a small amount of sensory data could improve predictions considerably.
- Creating a real-time probability scoring system for coffee quality to support business decisions.

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Appendix

- **Link to GitHub Repository:** [Insertlink]
- **Data Source:** <https://github.com/jldbc/coffee-quality-database>